

Enterococcus Intestinal Domination Is Associated With Increased Mortality in the Acute Leukemia Chemotherapy Population

Julia A. Messina,^{1,9} Chin Yee Tan,^{2,3} Yi Ren,⁴ Lauren Hill,⁵ Amy Bush,⁵ Meagan Lew,⁵ Tessa Andermann,⁶ Jonathan U. Peled,⁷ Antonio Gomes,⁸ Marcel R.M. van den Brink,⁷ Nelson J. Chao,^{4,5} Neeraj K. Surana,^{2,3,9,a} and Anthony D. Sung^{4,5,a}

¹Division of Infectious Diseases, Department of Medicine, Duke University, Durham, North Carolina, USA; ²Division of Infectious Diseases, Department of Pediatrics, Duke University, Durham, North Carolina, USA; ³Department of Molecular Genetics and Microbiology, Duke University, Durham, North Carolina, USA; ⁴Duke Cancer Institute, Duke University, Durham, North Carolina, USA; ⁵Division of Hematologic Malignancies and Cellular Therapy, Department of Medicine, Duke University, Durham, North Carolina, USA; ⁶Division of Infectious Diseases, Department of Medicine, University of North Carolina, Chapel Hill, North Carolina, USA; ⁷Adult Bone Marrow Transplantation Service, Memorial Sloan Kettering Cancer Center and Weill Cornell Medical College, New York, New York, USA; ⁸Sloan Kettering Institute, Memorial Sloan Kettering Cancer Center, New York, New York, USA; and ⁹Department of Immunology, Duke University, Durham, North Carolina, USA

Background. *Enterococcus* intestinal domination (EID), a state of dysbiosis wherein enterococci comprise $\geq 30\%$ abundance within the microbiota, has been associated with *Enterococcus* bacteremia, graft-versus-host disease, and mortality in the allogeneic hematopoietic stem cell transplant (allo HCT) population. The acute leukemia (AL) chemotherapy population includes patients receiving intensive chemotherapy, but not all patients go on to have an allo HCT. The impact of EID on outcomes including mortality in the AL chemotherapy population is unknown.

Methods. Microbiota composition from weekly stool samples was analyzed using 16S ribosomal RNA gene sequencing. Patients were analyzed in 2 cohorts: AL chemotherapy cohort and allo HCT cohort. α -diversity and richness were calculated. Kaplan-Meier analysis was used to analyze mortality.

Results. A total of 929 stool samples were collected from 139 patients. Both allo HCT and AL cohorts had a decline in α -diversity and richness over the course of treatment that tended not to return to baseline months later. EID was observed in at least 1 sample in 36% of allo HCT patients and 49% of AL patients. Patients with observed EID had lower α -diversity over time. Similar to the HCT cohort, AL patients with EID had reduced overall survival. We identified 4 other genera that were commonly found in $\geq 30\%$ relative abundance within the microbiota, but none were associated with mortality. In fact, in allo HCT, *Bacteroides* abundance $\geq 30\%$ was associated with improved survival.

Conclusions. EID is associated with increased all-cause mortality in HCT and AL cohorts. Unlike EID, relative abundance $\geq 30\%$ by other genera is not associated with increased all-cause mortality.

Keywords. microbiome; *Enterococcus*; intestinal domination; acute leukemia; hematopoietic stem cell transplant.

Patients with hematologic malignancies remain at disproportionately high risk for mortality despite advances in chemotherapy and hematopoietic stem cell transplantation (HCT) [1–3]. It is suspected that increased risk of mortality is, in part, due to intestinal domination, which commonly occurs with allogeneic HCT and acute leukemia intensive chemotherapy [4–7]. Intestinal domination has typically been defined as a state of dysbiosis wherein 1 bacterial genus comprises $>30\%$ relative abundance within the gut microbiota [4]. A previous study has demonstrated that *Enterococcus*, *Streptococcus*, and *Proteobacteria* are the most frequently encountered genera

with relative abundance $\geq 30\%$ and that *Enterococcus* domination is associated with a 9-fold increased risk for vancomycin-resistant *Enterococcus* (VRE) bloodstream infection (BSI), while *Proteobacteria* domination is linked with a 5-fold increased risk of gram-negative BSI [4, 8].

In addition to BSI, other adverse clinical outcomes have been demonstrated in association with *Enterococcus* intestinal domination (EID). We performed a multicenter, international cohort microbiome study of pre- and post-transplant longitudinal stool samples [9]. We found that intestinal domination occurred in a consistent manner across transplant centers and that *Enterococcus* was the most common dominating genus. Additionally, *Enterococcus* domination was associated with moderate to severe acute graft-versus-host disease (GVHD) and significant reduction in overall survival in the peri-transplant period (day 0 to +21), even after controlling for graft source, disease, conditioning intensity, gender, age, and comorbidity index [10]. It is unknown whether relative abundance $\geq 30\%$ of other genera has a similar deleterious impact on mortality.

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^aN. K. S. and A. D. S. contributed equally to this work.

Correspondence: Julia A. Messina, 315 Trent Drive, Room 253, Duke University, Durham, NC, 27705 (Julia.messina@duke.edu).

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Additionally, the impact of *Enterococcus* domination on mortality in nontransplant patients with hematologic malignancies is unknown.

Similar to the allogeneic HCT population, patients receiving intensive chemotherapy for acute leukemia have high rates of treatment-related complications and mortality, and there is growing interest in understanding the role of the gut microbiota in this population. In a prior study, it was found that the baseline microbiota composition is not significantly different between patients receiving chemotherapy for acute leukemia and patients undergoing allogeneic HCT [11]. In both groups, microbial diversity decreases over the course of therapy while domination events by bacteria such as *Enterococcus* (both acute leukemia and HCT cohorts) and *Lactobacillus* (acute leukemia cohort only) increase [11]. However, associations between these domination states and clinical outcomes have not been assessed.

Our primary objective in the present study was to investigate the association between EID and adverse clinical outcomes in patients receiving intensive chemotherapy for acute leukemia. Our secondary objectives were to compare adverse clinical outcomes associated with *Enterococcus* domination between the acute leukemia and allogeneic HCT populations and compare mortality between EID and relative abundance $\geq 30\%$ of other genera in the acute leukemia and allogeneic HCT populations.

METHODS

Patient Population

This prospective cohort study included 139 patients from whom weekly stool samples were collected from October 2013 through May 2019. Patients provided informed consent to participate in our biorepository, which was approved by the Duke University Institutional Review Board. Patients were included in the present study if they had ≥ 4 weekly stool samples during the study period. The allogeneic HCT cohort included patients undergoing allogeneic HCT for hematologic malignancies, whose stool collections occurred once during the pre-transplant period (starting at day -30 through day 0), weekly through day $+30$ and then at days $+45$, day $+90$, $+180$, and annually post-transplant. For the acute leukemia chemotherapy cohort, stool samples were collected weekly during intensive induction or salvage chemotherapy. Two patients in this cohort went on to receive an allogeneic HCT but were only analyzed in the acute leukemia cohort for the purposes of this study.

Stool Sample Collection and Sequencing

The samples were placed in a refrigerator at 4°C immediately upon collection, followed by transport from the clinical units within 24 hours of collection and storage in a -80°C freezer without additional substrates or preservatives. Polymerase chain reaction (PCR) amplification of genomic 16S ribosomal RNA V4–V5 regions was performed and sequenced on the Illumina

MiSeq with 250bp paired-end reads as previously described [5, 9, 12]. Raw sequence data were uploaded into the Sequence Read Archive at the National Center for Biotechnology Information (accession PRJNA772002).

Clinical Outcomes

Clinical outcomes such as date of last follow-up and mortality were evaluated starting from transplant date (allogeneic HCT) or chemotherapy start date corresponding with study entry (acute leukemia chemotherapy) through 31 May 2020. We investigated the following types of mortality: all-cause mortality, death due to any cause within 1 year from chemotherapy or transplant start date; relapse-related mortality, death from relapsed/refractory disease; and treatment-related mortality, death due to complications of therapy such as infection or end-organ failure without relapsed/refractory disease. We assessed clinical outcomes related to morbidity from therapies within a 1-year time frame of stool collection for individual patients. BSI was defined by Centers for Disease Control and Prevention criteria as ≥ 1 positive blood culture for bacterial pathogens except for common skin contaminants such as coagulase-negative staphylococci that require ≥ 2 positive blood cultures on the same day or consecutive days [13]. *Clostridioides difficile* colitis was defined by the presence of diarrhea (≥ 3 loose or liquid stools within 24 hours) and a positive PCR test for *C. difficile* toxins. Mucositis and GVHD were determined by chart review as recorded by the treating medical providers. Data for positive VRE rectal swab were only analyzed for the HCT cohort as these patients undergo weekly surveillance testing through day $+30$ post-transplant (in contrast, the acute leukemia cohort was not routinely screened for VRE colonization).

Bioinformatics Analysis

Demultiplexed sequencing reads were processed with the DADA2 pipeline (version 1.16.0, R version 4.0.0) on a per sequencing run basis [14]. DADA2 outputs from all runs were merged, and taxonomic assignment was performed using the Silva SSU taxonomic training dataset (version 138) with taxonomic assignment to the species level where possible. Taxonomic tables were agglomerated to the genus level, and domination events defined by relative abundance of a given genus ≥ 0.3 were computed for each sample. The domination threshold of ≥ 0.3 was selected a priori based on the findings from prior studies [4, 9, 10, 15]. This threshold was also used to designate high vs low relative abundance of *Bacteroides* based on the threshold of <0.3 set by the Human Microbiome Project [16]. α -diversity was determined using the inverse Simpson index. α -diversity and taxonomic richness were generated and visualized using phyloseq (version 1.30.0) [17]. Differences in α -diversity between cohorts were compared using the Mann-Whitney *U* test.

Statistical Analysis

Categorical variables were summarized by proportions and compared using the Fisher exact test. Continuous variables were summarized by median and interquartile range (IQR) and compared using the *t* test. We compared rates of GVHD (acute, chronic, and gut; all grades/stages included), relapsed disease, BSI, mucositis, *C. difficile* colitis, and mortality between patients with and without EID. The duration of follow-up was determined relative to the start date of chemotherapy for leukemia or the transplant date for the HCT. We used Kaplan-Meier survival curves to visualize overall and relapse-free survival and compared the cohorts using the log-rank test. To analyze treatment-related mortality, we used Fine-Gray subdistributional hazard regression where deaths were considered as competing risk. A *P* value <.05 was considered statistically significant.

RESULTS

A total of 139 patients were included in this study with 929 stool samples (681 from 98 allogeneic HCT patients and 248 stool samples from 41 acute leukemia patients) undergoing 16S rRNA gene sequencing. Fifty-five patients (40%) had at least 1 stool sample that met the definition for EID during the study. Figure 1A depicts a comparison in α -diversity between the acute leukemia and HCT cohorts. The HCT cohort has significantly lower α -diversity among aggregate stool samples compared with the acute leukemia cohort.

Figure 1B presents a decline in α -diversity over the course of therapy in both the allogeneic HCT and acute leukemia cohorts, although the timing of the nadir in diversity is different. The HCT cohort tends to have the lowest reduction in α -diversity and richness around day +14 post-HCT, while the nadir for the acute leukemia cohort occurs later in

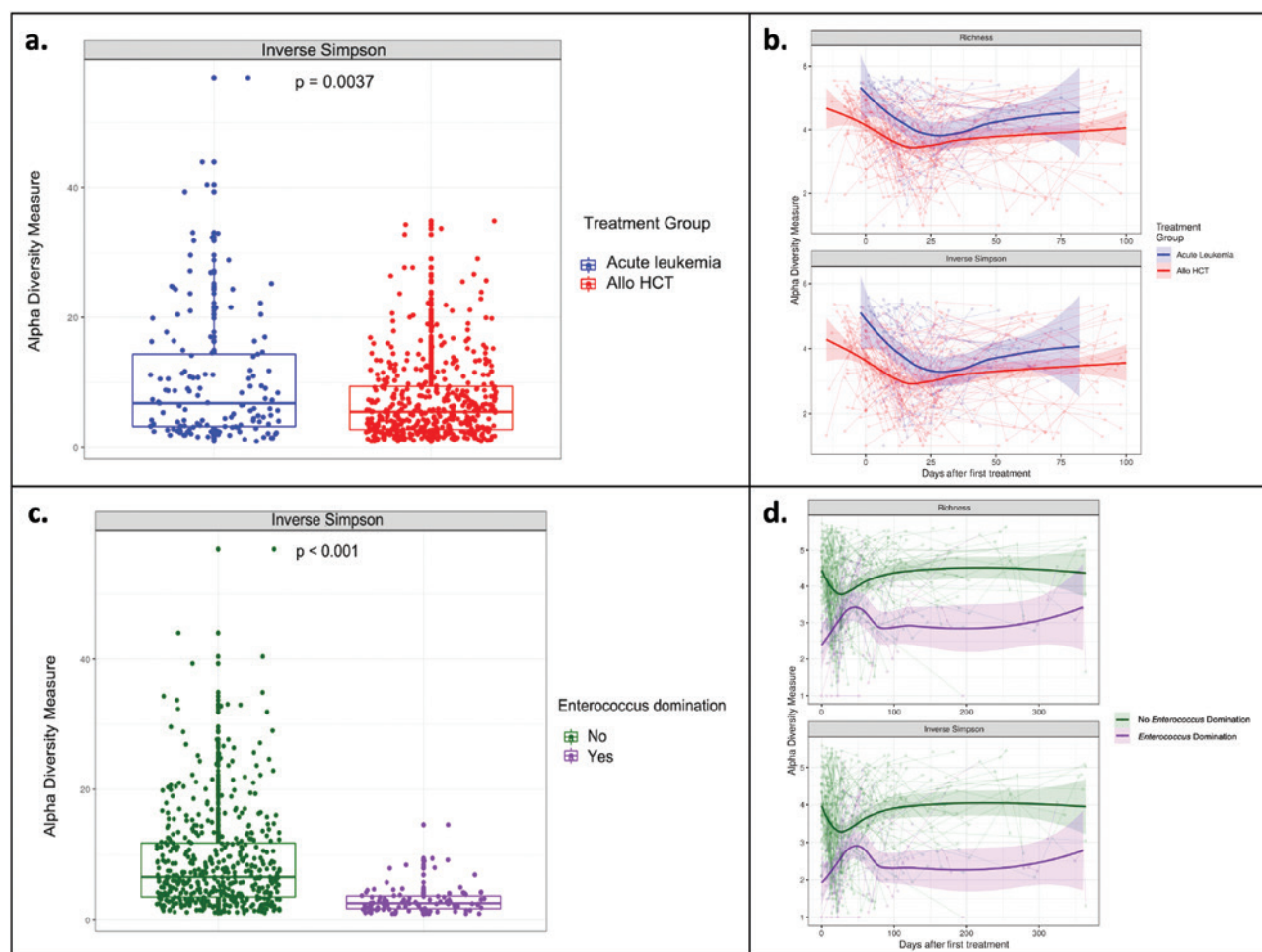


Figure 1. A, Differences in α -diversity by cohorts (allogeneic HCT and acute leukemia chemotherapy groups are combined for analysis). B, Richness and α -diversity comparison among cohorts over time (first treatment for acute leukemia patients = start date of chemotherapy; for allogeneic HCT, day 0 = day of stem cell infusion). C, Differences in α -diversity by *Enterococcus* intestinal domination (allogeneic HCT and acute leukemia chemotherapy groups are combined for analysis). D, Richness and α -diversity by *Enterococcus* intestinal domination over time (allogeneic HCT and acute leukemia cohorts are combined for analysis). First treatment for acute leukemia patients = start date of chemotherapy; for allogeneic HCT, day 0 is day of stem cell infusion). Abbreviation: HCT, hematopoietic stem cell.

treatment, around day 25 following the start of chemotherapy. Additionally, each cohort demonstrates a subsequent gradual increase in α -diversity and richness over time, though it does not normalize to baseline α -diversities in either group. Although this trend of decreased α -diversity and richness is present in each cohort, α -diversity and richness are higher overall in the acute leukemia cohort.

Figure 1C depicts α -diversity and richness stratified by the presence or absence of EID at any point during therapy. Patients with at least 1 episode of *Enterococcus* domination have significantly lower α -diversity of the gut microbiota over the course of therapy (Figure 1D; $P < .001$).

Ninety-eight patients from the allogeneic HCT cohort were evaluated, and 35 (36%) had at least 1 stool sample with EID. The median duration between transplant date and the first stool sample with *Enterococcus* domination was 22 days (IQR, 6.75–84.25). Of note, 2 patients already had *Enterococcus* domination present in the pre-transplant stool specimen. Baseline demographics, depicted in Table 1, were comparable between groups. Regarding clinical outcomes among the HCT cohort, patients with *Enterococcus* domination were more likely to have at least 1 BSI, develop VRE rectal colonization, and have *C. difficile* colitis (Table 3). Patients with *Enterococcus* domination trended toward having more gut GVHD, although

rates of acute and chronic GVHD were not significantly different between groups. Patients with *Enterococcus* domination also had reduced overall survival (Figure 2A), increased treatment-related mortality (Supplementary Figure 1A), and increased relapse-related mortality (Supplementary Figure 2A).

Among the 41 patients from the acute leukemia chemotherapy cohort, 20 (49%) had at least 1 stool sample with EID. The median duration between chemotherapy start date (day 0) and the first stool sample with *Enterococcus* domination was 28 days (IQR, 14.5–41.75). Baseline demographics were comparable between patients with and without *Enterococcus* domination, although patients with *Enterococcus* domination tended to be younger (Table 2). Additionally, there was no difference between the 2 cohorts in the number of patients undergoing induction vs reinduction/salvage chemotherapy. Within this cohort, patients with *Enterococcus* domination had more BSI, *C. difficile* infections, and mucositis, though none of these associations reached statistical significance (Table 4).

Surprisingly, of the 11 patients with *Enterococcus* BSI (3 in the allogeneic HCT cohort and 8 in the acute leukemia cohort), only 3 patients had *Enterococcus* domination in their fecal microbiota preceding their BSI (range, 1–20 days from domination event to BSI).

Table 1. Allogeneic Hematopoietic Stem Cell Transplant Cohort Demographics

Characteristic	<i>Enterococcus</i> Intestinal Domination, N = 35 (36%)	No <i>Enterococcus</i> Intestinal Domination, N = 63 (64%)	P Value
Age, ^a years	52 (35–63)	55 (45–63)	.12
Underlying disease (%)			.13
AML	15 (43)	16 (25)	
MDS	5 (14)	15 (24)	
Other	15 (43)	32 (51)	
Female gender (%)	12 (34)	17 (27)	.49
Race (%)			.03
White	26 (74)	55 (87)	
Black	7 (20)	3 (5)	
Other	2 (6)	5 (8)	
Non-Hispanic or Latino ethnicity (%)	35 (100)	58 (92)	.53
Charlson comorbidity index ^a	3 (2–4)	2 (2–3)	.20
Conditioning (%)			1.00
Myeloablative	33 (94)	58 (92)	
Nonmyeloablative	2 (6)	4 (6)	
Reduced intensity	0	1 (2)	
HLA match (%)			.97
Matched related	11 (31)	21 (33)	
Matched unrelated	22 (63)	39 (62)	
Haploidentical	2 (6)	3 (5)	
Stem cell source (%)			.73
Bone marrow	26 (74)	42 (67)	
Peripheral blood cells	5 (14)	11 (17)	
Cord blood cells	4 (11)	10 (16)	
Median number of stool samples analyzed per patient ^a	6 (5–10)	6 (5–8)	.22

Continuous variables compared using the *t* test, and categorical variables compared using the Fisher exact test.

^aSummarized by median and interquartile range.

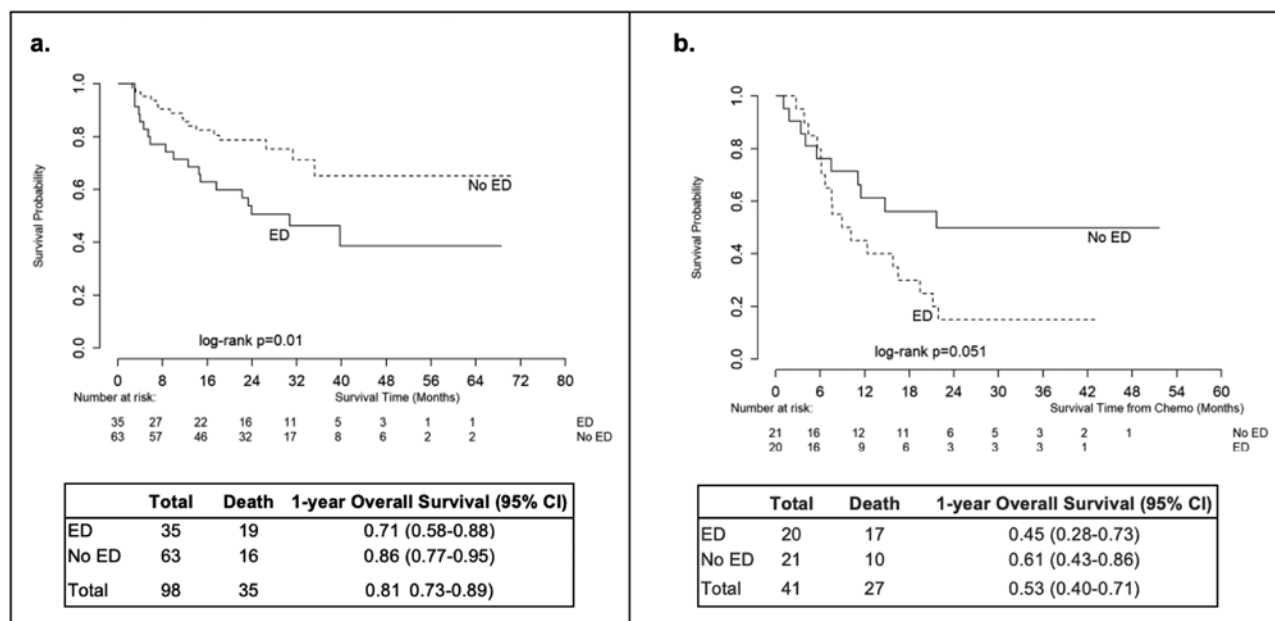


Figure 2. A, Overall survival for allogeneic transplant cohort based on *Enterococcus* domination. B, Overall survival for acute leukemia cohort based on *Enterococcus* domination. Abbreviations: CI, confidence interval; ED, *Enterococcus* domination.

Two patients did not have stool samples collected prior to the BSI but did have *Enterococcus* domination in samples collected shortly after diagnosis of their BSI, 4 patients developed *Enterococcus* domination after their BSI was diagnosed (range, 2–171 days), and 2 patients never developed *Enterococcus* domination.

Figure 2B depicts Kaplan-Meier analysis of overall survival for the acute leukemia chemotherapy cohort. Median duration

of follow-up among patients who survived median 2.8 years (IQR, 1.8–3.4; compared with 2.3 years [IQR, 1.7–3.2]; $P = .38$) among allogeneic HCT patients who survived). Patients with *Enterococcus* domination had reduced overall survival although this result did not quite reach statistical significance ($P = .051$). There was no association between *Enterococcus* domination and reduced treatment-related survival, but this analysis was limited

Table 2. Acute Leukemia Chemotherapy Cohort Demographics

Characteristic	<i>Enterococcus</i> Intestinal Domination, N = 20 (49%)	No <i>Enterococcus</i> Intestinal Domination, N = 21 (51%)	P Value
Age, years ^a	54.5 (43–63.5)	60 (54–70)	.05
Underlying disease (%)			
AML	19 (95)	18 (86)	.32
ALL	1 (5)	3 (14)	
Stool collection period (%)			
Induction	14 (70)	16 (76)	.65
Reinduction/Salvage	6 (30)	5 (24)	
Female gender (%)	13 (65)	10 (48)	.35
Race (%)			.84
White	15 (75)	14 (67)	
Black	3 (15)	5 (24)	
Other	2 (10)	3 (14)	
Non-Hispanic or Latino ethnicity (%)	20 (100)	21 (100)	1.00
Charlson comorbidity index ^a	3.5 (2–5)	3 (3–5)	.31
Chemotherapy regimen at time of stool collection (%)			1.00
7 + 3 ^b	17 (85)	18 (86)	
Other	3 (15)	3 (14)	
Median ^b number of stool samples analyzed per patient ^a	6.5 (4–8.5)	5 (4–7)	.30

Continuous variables compared using the *t* test, and categorical variables compared using the Fisher exact test.

^aSummarized by median and interquartile range.

^bCytarabine + doxorubicin or variants of these drugs.

Table 3. Allogeneic Hematopoietic Stem Cell Transplant Cohort Outcomes

Outcome	<i>Enterococcus</i> Intestinal Domination, N = 35 (36%)	No <i>Enterococcus</i> Intestinal Domination, N = 63 (64%)	PValue
Acute GVHD (%)	14 (40)	30 (48)	.53
Chronic GVHD (%)	18 (51)	29 (46)	.68
Gut GVHD (%)	19 (54)	23 (37)	.09
Any bloodstream infection (%)	22 (63)	22 (35)	.01
	34 events	28 events	
Coagulase-negative <i>Staphylococcus</i>	6	6	
<i>Enterococcus</i>	4	0	
Viridans <i>Streptococcus</i>	4	6	
<i>Escherichia coli</i>	3	5	
<i>Klebsiella</i>	3	2	
<i>Pseudomonas</i>	2	1	
<i>Stenotrophomonas</i>	2	1	
<i>Staphylococcus aureus</i>	1	2	
<i>Citrobacter</i>	1	1	
<i>Enterobacter</i>	1	1	
<i>Candida</i>	1	0	
Other	5	6	
<i>Clostridioides difficile</i> colitis (%)	12 (34)	10 (16)	.04
Mucositis (%)	29 (83)	40 (64)	.06
Positive vancomycin-resistant <i>Enterococcus</i> rectal swab ^b (%)	21 (60)	16 (25)	.001
Relapsed disease post-transplant (%)	10 (29)	17 (27)	1.00

Continuous variables compared using the *t* test, and categorical variables compared using the Fisher exact test.

Abbreviation: GVHD, graft versus host disease.

^aSummarized by median and interquartile range.

^bOutcome assessed over time at any time point.

by only 3 patients within the acute leukemia cohort suffering from treatment-related mortality (Supplementary Figure 1B). Patients with *Enterococcus* domination did have higher rates of relapse-related mortality (Supplementary Figure 2B).

We observed multiple instances of genera other than *Enterococcus* having a relative abundance $\geq 30\%$. Figure 3

depicts the proportion of patients within each treatment group (allogeneic HCT and acute leukemia induction chemotherapy) who experienced at least 1 such event. While some of these genera (*Akkermansia*, *Lactobacillus*, and *Streptococcus*) are uncommonly present in the fecal microbiome of healthy individuals (mean relative abundance $<1\%$ for all), *Bacteroides* and

Table 4. Acute Leukemia Chemotherapy Cohort Outcomes

Outcome	<i>Enterococcus</i> Intestinal Domination, N = 20 (49%)	No <i>Enterococcus</i> Intestinal Domination, N = 21 (51%)	PValue
Any bloodstream infection (%)	15 (70)	11 (52)	.34
	27 Events	16 Events	
Coagulase-negative <i>Staphylococcus</i>	7	2	
<i>Enterococcus</i>	6	2	
Viridans <i>Streptococcus</i>	3	2	
<i>Escherichia coli</i>	2	1	
<i>Klebsiella</i>	1	0	
<i>Pseudomonas</i>	1	2	
<i>Staphylococcus aureus</i>	1	1	
<i>Citrobacter</i>	0	1	
Other	6	5	
<i>Clostridioides difficile</i> colitis (%)	9 (45)	5 (24)	.20
Mucositis (%)	10 (50)	7 (33)	.35
Relapsed disease (%)	15 (75)	10 (48)	.11

Continuous variables compared using the *t* test, and categorical variables compared using the Fisher exact test.

^aSummarized by median and interquartile range.

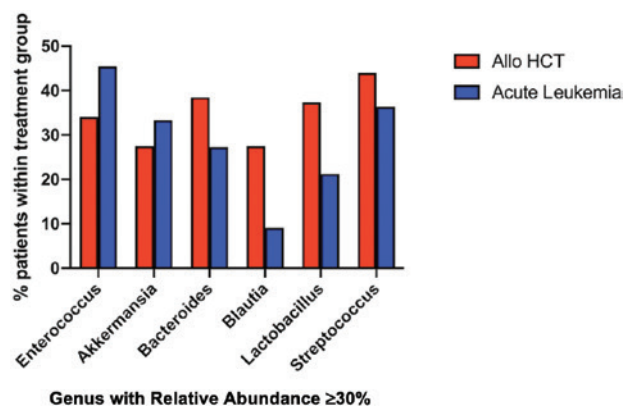


Figure 3. Proportion of patients with relative abundance $\geq 30\%$ by genera other than *Enterococcus*. Abbreviation: HCT, hematopoietic stem cell.

Blautia are present at higher mean relative abundances (approximately 50% and approximately 5%, respectively) [16, 18]. Of note, a *Bacteroides* relative abundance of $\leq 30\%$ represents the bottom quartile of healthy patients. Although *Enterococcus* was the most common dominating genus in the acute leukemia cohort, affecting 49% of patients, it was only present in 36% of the allogeneic HCT cohort where *Streptococcus* was the most commonly identified genus with a relative abundance $\geq 30\%$ (42% of patients). We also observed that 18 patients (6 patients in the chemotherapy cohort and 12 in the allogeneic HCT cohort) experienced codomination events, where 2 of the 5 most common genera were both present at $\geq 30\%$ relative abundance. *Enterococcus* was the most common genus to be present in codomination (occurring in 7 patients) and was most commonly paired with *Lactobacillus* (4 patients). Additionally, 5 patients had ≥ 1 stool sample with codomination.

To determine whether genera other than *Enterococcus* were associated with all-cause

mortality, we performed a Kaplan-Meier analysis (Figure 4; Supplementary Figure 3A–F). None of the other genera were associated with an increased rate of all-cause mortality. However, *Bacteroides* relative abundance $\geq 30\%$ was associated with reduced all-cause mortality in the allogeneic HCT cohort ($P = .08$; Figure 4A) with a trend toward being protective in the chemotherapy cohort ($P = .12$; Figure 4B), although these did not reach statistical significance.

DISCUSSION

Consistent with previous work, we found that EID is associated with increased mortality in the allogeneic HCT cohort. In addition, we identified additional adverse clinical outcomes in the allogeneic HCT cohort associated with *Enterococcus* domination, including the development of any BSI and *C. difficile* colitis. It is interesting that even though the conditioning and chemotherapy regimens for HCT and acute leukemia use different cytotoxic agents, those patients receiving intensive chemotherapy also commonly develop *Enterococcus* domination, which is similarly associated with increased mortality.

We identified 5 other bacterial genera that are the most common causes of high abundance using the $\geq 30\%$ relative abundance cutoff, including *Akkermansia*, *Bacteroides*, *Blautia*, *Lactobacillus*, and *Streptococcus*. Unlike *Enterococcus* domination, relative abundance $\geq 30\%$ with these other genera is not associated with an increased all-cause mortality. In fact, *Bacteroides* relative abundance $\geq 30\%$ in the allogeneic HCT cohort appears to be associated with higher overall survival.

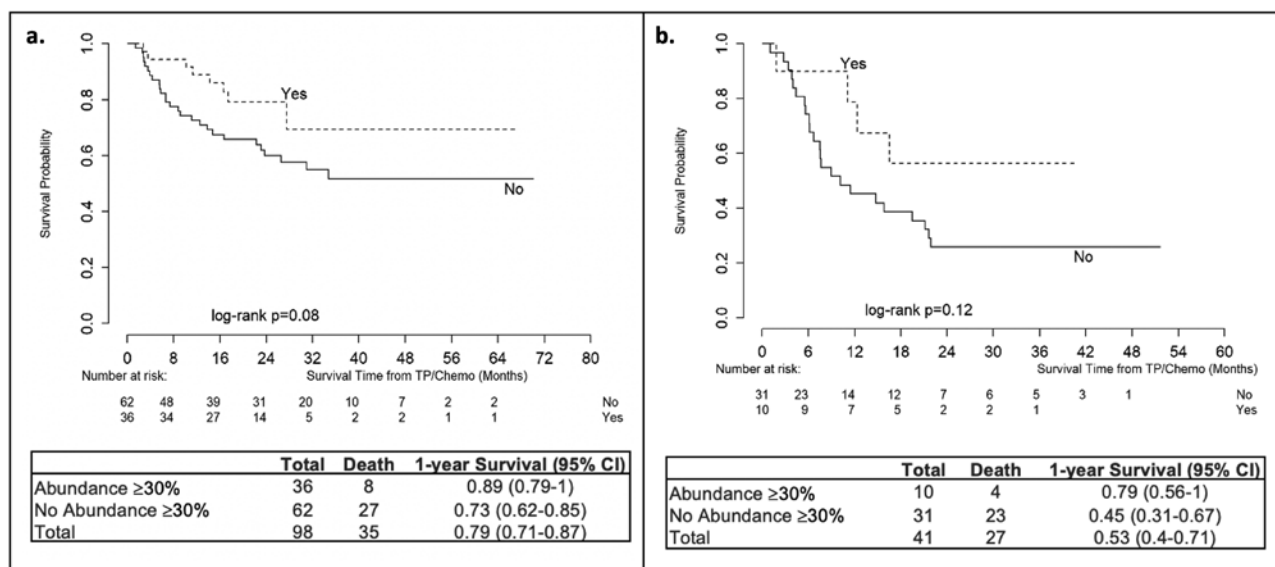


Figure 4. A, All-cause mortality and *Bacteroides* relative abundance $\geq 30\%$ and overall survival in allogeneic TP cohort. B, All-cause mortality and *Bacteroides* relative abundance $\geq 30\%$ and overall survival in acute leukemia cohort. Abbreviations: CI, confidence interval; TP, transplant.

Our findings are consistent with previous work comparing the fecal microbiota of patients undergoing chemotherapy for acute leukemia with patients undergoing allogeneic HCT. Despite differences in treatments, both cohorts experience reduced diversity following treatment and the development of domination events [19]. In a previous study, investigators also noted the presence of codomination in an acute leukemia cohort, with *Enterococcus* and *Lactobacillus* codomination occurring in 17% of patients, and found that use of third-generation or higher cephalosporins and metronidazole or clindamycin increased the risk for *Enterococcus* domination. The investigators did not observe events of *Streptococcus* relative abundance $\geq 30\%$. Of note, *Streptococcus* was the most common genus with relative abundance $\geq 30\%$ in the HCT cohort in our study. As the investigators note, these differences in dominating genera may be due to differences in antimicrobial prophylaxis and febrile neutropenia protocols. Future multicenter studies that control for these differences will be critical to further understanding risk factors for *Enterococcus* domination.

In the present study, we report a number of differences in characterizing *Enterococcus* domination between the acute leukemia and allogeneic HCT cohorts. First, *Enterococcus* domination is more common in the acute leukemia cohort (49%) compared with the HCT cohort (36%). Second, all-cause mortality in patients with *Enterococcus* domination tends to occur earlier (within the first year after therapy) in the acute leukemia cohort compared with the HCT cohort (within the first 30 days post-transplant).

We also found that BSIs are more common in patients with *Enterococcus* domination in both cohorts. This finding was statistically significant in the allogeneic HCT cohort but did not reach statistical significance in the acute leukemia cohort. BSI in patients with *Enterococcus* domination may be more common because these patients have a lower α -diversity, which is known to be associated with poor outcomes [5, 9]. It is unclear whether it is the presence of *Enterococcus* specifically or the loss of other commensal bacteria that leads to this increase in BSI.

In the allogeneic HCT cohort, the finding that VRE rectal colonization is associated with *Enterococcus* domination is significant and suggests that enterococci that dominate the gut microbiota may be more likely to be vancomycin resistant. In a prior study of *Enterococcus* domination in allogeneic HCT patients, 92% of patients with *Enterococcus* domination had detectable expression of the *vanA* gene, a finding that suggests the presence of VRE [4]. A prior study found that allogeneic HCT patients with GVHD tended to have increased *Enterococcus* abundance, and enterococcal PCR confirmed the presence of *Enterococcus faecalis* and *Enterococcus faecium* but not other enterococcal species [7]. Based on these varied findings, further work is needed to determine which enterococcal genetic lineages and genes contribute to *Enterococcus* domination.

A notable finding from the present study is that *Bacteroides* relative abundance $\geq 30\%$ in the allogeneic HCT cohort is associated with increased survival, with a trend toward increased survival in the acute leukemia cohort. *Bacteroides* are obligate anaerobes typically present in high proportions within the healthy human gut microbiota and may be important in maintaining colonization resistance to pathogenic bacteria [20, 21]. *Bacteroides* species have also been identified in murine models as playing critical roles in the antitumor effect of CTLA-4 inhibitor therapy and can boost the effects of chemotherapeutic agents [22, 23]. Further study is needed as to how *Bacteroides* abundance, and which species in particular, is associated with differences in survival [22]. Further study is also needed to determine whether increased *Bacteroides* abundance confers colonization resistance by more pathogenic bacteria and thus reduces treatment-related mortality as well as what impact it has on the graft-versus-tumor effect of HCT and potential prevention of relapse-related mortality.

The term “domination” may not be correct for all organisms as it can imply pathogenicity leading to adverse clinical outcomes. For example, *Bacteroides* can be found at relatively high abundances in healthy populations [16]. Future study is needed into the roles of these additional genera within microbial networks to better understand the clinical significance of high- and low-abundance states in the hematologic malignancy population.

The present study is limited by its single-center design and lack of pre-transplant/chemotherapy stool sample for every patient to effectively characterize changes to the gut microbiota composition over time. Additionally, the small sample size of the acute leukemia cohort limited analysis of other important factors such as comorbid conditions and antimicrobial therapy in establishing a causal relationship between *Enterococcus* domination and mortality.

In conclusion, we found that EID is associated with increased risk for all-cause mortality in both the acute leukemia and allogeneic HCT cohorts. Its relationship with other clinical outcomes differs between the 2 patient populations, which may imply different mechanisms for pathogenesis. We also found that not all states of relative abundance $\geq 30\%$ are detrimental and that *Bacteroides* abundance in the allogeneic HCT cohort may be protective against mortality. Future multicenter studies are needed to further understand the impact of *Enterococcus* domination in the acute leukemia population and how *Bacteroides* abundance leads to improved outcomes in the allogeneic HCT population.

Supplementary Data

Supplementary materials are available at *Clinical Infectious Diseases* online. Consisting of data provided by the authors to benefit the reader, the posted materials are not copyedited and are the sole responsibility of the authors, so questions or comments should be addressed to the corresponding author.

Notes

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